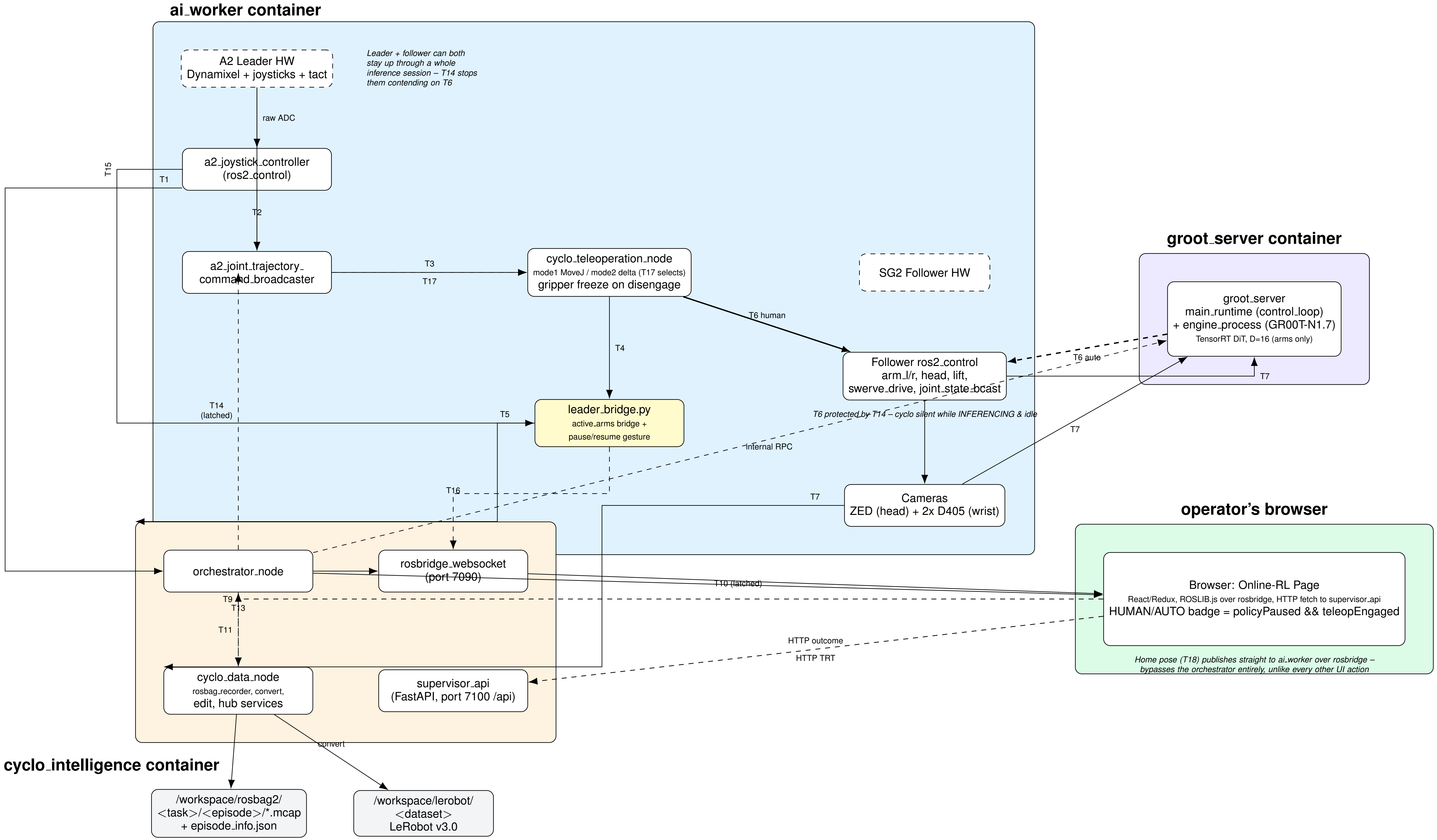
HG-DAgger / Online-RL System Architecture — FFW SG2

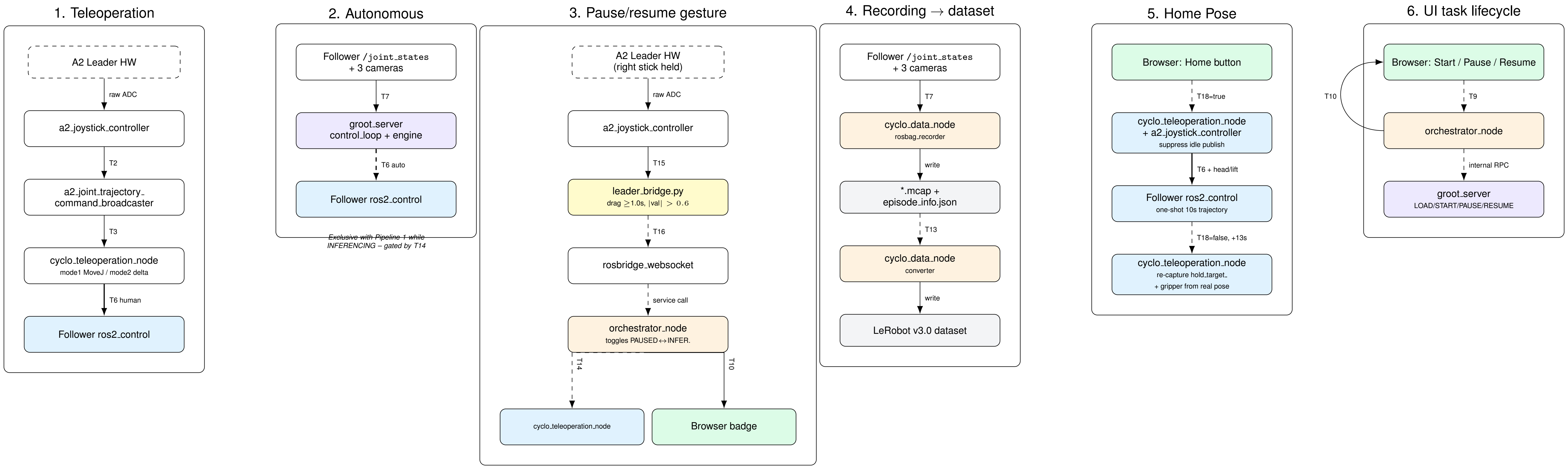
ai_worker (leader + follower hardware) · cyclo_intelligence (orchestration) · groot_server (policy)

Edge labels are topic/service IDs – p.2 walks each core data path step by step, p.3+ defines every ID



Core Pipelines — Step by Step

Each box is one node/process; each arrow is one topic or service call, read top to bottom. IDs are defined on p.3+



Topic / Service Legend

ID	Kind	Name	Type	Note
T1	PUB/SUB	`.../tact_trigger`	`std_msgs/String`	"left"/"right" tap. Ignored for recording (`record_triggers_enabled=false`).
T2	PUB/SUB	`.../joystick_command`	`robotis_interfaces` `/TeleoperationCommand`	ENABLE/DISABLE/TOGGLE per arm. `a2_joystick_controller` → broadcaster.
T3	PUB/SUB	`.../raw_joint_trajectory` (internal)	`trajectory_msgs/JointTrajectory`	Broadcaster's leader reference → `cyclo_teleoperation_node`.
T17	PUB/SUB	`.../control_command`	`robotis_interfaces/ControlModeCommand`	Selects mode (1=absolute, 2=delta, default) + enabled arms.
T4	PUB/SUB	`.../control_status`	`robotis_interfaces/ControlModeStatus`	`active_arms`/mode/sync. Not decodable in `cyclo_intelligence` – excluded from recording.
T5	PUB/SUB latched	`.../active_arms_str`	`std_msgs/String`	Decodable translation of T4. UI badge subscribes here.
T15	PUB/SUB	`.../sensorxel_joy_values`	`std_msgs/Float64MultiArray`	Raw joystick axes. `leader_bridge.py` watches R-stick Y for the pause gesture.
T16	PUB/SUB	`.../toggle_pause_request`	`std_msgs/Empty`	Fires once on a ≥1.0s drag-hold. Orchestrator decides Pause vs Resume.
T6	PUB/SUB	`.../joint_trajectory_command_broadcaster_{l,r}/...`	`trajectory_msgs/JointTrajectory`	Arm commands. 3 possible publishers (teleop/policy/Home); guarded by T14.
T18	PUB/SUB latched	`.../home_in_progress`	`std_msgs/Bool`	Bypasses orchestrator. One-shot Home trajectory on T6; falling edge re-captures gripper.
T14	PUB/SUB latched	`/task/policy_active`	`std_msgs/Bool`	Mirrors T10's INFERENCING for ai_worker. Gates T6's idle publish.
	PUB/SUB	`.../joystick_controller_{l,r}/joint_trajectory`	`trajectory_msgs/JointTrajectory`	Head/lift jog. Frozen during policy control.
	PUB/SUB	`/cmd_vel`	`geometry_msgs/Twist`	Mobile base. Frozen during policy control.
T7	PUB/SUB	`/joint_states` + camera topics	`sensor_msgs/JointState,CompressedImage`	Follower feedback + 3 cameras. Feeds broadcaster/policy/recorder/preview.
T9	SERVICE	`/task/command`	`SendCommand.srv`	Start/Pause/Resume/Clear policy; Start/Stop/Cancel recording.
T10	PUB/SUB latched	`/task/inference_status`	`interfaces/InferenceStatus`	READY/LOADING/INFERENCING/PAUSED phase.
T11	SERVICE	`/data/recording`	`RecordingCommand.srv`	Recording lifecycle: orchestrator → `cyclo_data`.
T13	SERVICE	`/data/convert(/status)`	`srv`	LeRobot v3.0 conversion. Triggered separately from Save.
	HTTP	`/api/recording/outcome`	REST (supervisor_api)	Writes `outcome+success_frame` to `episode_info.json`.
	HTTP	`/api/backends/groot/trt/*`	REST (supervisor_api)	TensorRT engine status/build.

Mode 2 (`AiWorkerElbowUpLeaderMode`, see T17): anchor-capture on engage, `goal = follower_anchor + (leader_now -- leader_anchor)` – zero jump even if leader and follower drifted apart. Mode 1 relays absolute leader positions and does jump.

Per-frame dataset columns (converter output)

Column	dtype	Derivation
`intervention`	int64	Causal fill of T10 onto the frame grid: `INFERENCING`→0, `PAUSED`→1, else -1. Frames during leader slow-start retargeting (broadcaster's `time_from_start` $\neq 0$) also forced to -1 – that motion is the clutch, not a human command. **Currently only the -1/0 half is reliable**: T4 (needed for the 1-vs--1 distinction) is not recorded (see T4 note); re-wiring the converter to read T5 instead is pending.
`sample_weight`	float32	1.0 by default. Every **autonomous** (`intervention`=0) frame within `_PRE_PAUSE_RAMP_SECONDS` (2.0s) immediately before a `PAUSED` onset ramps linearly $1.0 \rightarrow 0.0$ at the pause instant; human and excluded frames are untouched. Matches the real-robot DAgger weighting in Larchenko's LeHome Challenge solution (arXiv:2606.27163 §9.10) – don't reinforce the exact moves that led into a takeover. Their window was 5s; ours is shorter (2s) since we have no sim-replay bucket diluting the real-data signal the way theirs did.
`reward`	int64	1 from `success_frame` onward on `outcome=SUCCESS` episodes, else 0.
`done`	bool	True on the final frame of the episode only.

Diagram reflects verified state as of this session: gripper-freeze confirmed working; badge subscription fixed (orphaned rosbridge on port 7090 killed, fresh process now serving); T4 recording reverted after crashing the launch group; T5/T15/T16 all live in `leader_bridge.py`, still a standalone script, not packaged into any launch file. T14 (idle-publish fix) and T16 (pause/resume gesture) built this session – T14 needs `colcon build --packages-select cyclo_teleoperation` + an orchestrator restart to take effect; T16's right-stick axis index was derived from source reading with the leader down, not confirmed live yet.